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Linear Differential Equation

→ Linear differential equation :

Linear differential equations are those in which the dependent variable and its derivative occur only in the first degree and not multiplied together.

The general linear differential equation of the nth order is of the form :

$$\frac{d^n y}{dx^n} + P_1 \frac{d^{n-1} y}{dx^{n-1}} + P_2 \frac{d^{n-2} y}{dx^{n-2}} + \cdots + P_n y = X.$$

where, $P_1, P_2, \ldots P_n$ and X are function of x only.

Linear differential equation with constant co-efficients are in the form of :

$$\frac{d^n y}{dx^n} + k_1 \frac{d^{n-1} y}{dx^{n-1}} + k_2 \frac{d^{n-2} y}{dx^{n-2}} + \cdots + k_n y = X$$

where, $k_1, k_2, \ldots k_n$ are constants. Such equations are most important in the study of electro-mechanical vibration and other engineering problems.

→ Operator D :-

Denoting $\cdot \frac{d}{dx}$ by D

$\cdot \frac{d^2}{dx^2}$ by D^2

$\cdot \frac{d^3}{dx^n}$ by D^3

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$$\therefore \frac{dy}{dx} = Dy$$

$$\frac{d^2y}{dx^2} = D^2 y$$

$$\frac{d^3y}{dx^3} = D^3 y$$

$$\frac{d^n y}{dx^n} = D^n y \qquad \left[\frac{d^n}{dx^n} = D^n \right]$$

Eg;

We can write;

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - 3y = D^2y - 2Dy - 3y$$

$$= (D^2 - 2D - 3)y$$

$$= (D^2 - 3D + D - 3)y$$

$$= (D(D-3) + 1(D-3))y$$

$$= (D-3)(D+1)y$$

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→ Rules for finding the complementary function :-

→ Auxiliary equation (A.E) → roots of linear differential equation.

$$(D^n + k_1 D^{n-1} + k_2 D^{n-2} + \cdots k_n) y = 0$$

$$(D - m_1)(D - m_2) \cdots (D - m_n) = 0$$

let, $m_1, m_2, \cdots m_n$ = roots.

Eg; Find Auxiliary equation (A.E) of $(D^2 - 2D - 3)y = 0$

$$(D^2 - 2D - 3)y = 0$$

putting $m = D$

A.E ⇒ $m^2 - 2m - 3 = 0$

$(m - 3)(m + 1) = 0$

$m = 3$ or $m = -1$

∴ A.E is $m = 3, -1$

→ Rule for Complementary Function (CF) :-

Case - I ; If all roots are real and different :

i.e; $m_1, m_2, \cdots m_n$, all are real and

$m_1 \neq m_2 \neq m_3 \cdots \neq m_n$.

C.F ⇒ $y = C_1 e^{m_1 x} + C_2 e^{m_2 x} + \cdots C_n e^{m_n x}$

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Q.1. Solve;

$$\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = 0.$$

⇒ Given;

$$\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = 0$$

$$(D^2 + 5D + 6)y = 0$$

putting $m = D$

A.E ⇒ $m^2 + 5m + 6 = 0$

$\Rightarrow m^2 + 3m + 2m + 6 = 0$

$\Rightarrow m(m+3) + 2(m+3) = 0$

$\Rightarrow (m+3)(m+2)$

$\Rightarrow m = -3, -2$

C.F ⇒ $y = C_1 e^{-3x} + C_2 e^{-2x}$ $\left[C_1 e^{m_1 x} + C_2 e^{m_2 x}\right]$

Q.2. Solve;

$$y'' - 4y' - 5y = 0$$

⇒ Given;

$\Rightarrow y'' - 4y' - 5y = 0$

$\Rightarrow (D^2 - 4D - 5)y = 0$

putting $m = D$

A.E ⇒ $m^2 - 4m - 5 = 0$

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$\Rightarrow m^2 - 5m + m - 5 = 0$

$\Rightarrow m(m-5) + 1(m-5) = 0$

$\Rightarrow (m-5)(m+1) = 0$

$\Rightarrow m = 5, -1$

C.F $\Rightarrow y = C_1 e^{5x} + C_2 e^{-x}$

Q.2. Solve;

$$y'' - 4y' + 5y = 0$$

$\Rightarrow$ Given;

$$y'' - 4y' + 5y = 0$$

$$\Rightarrow (D^2 - 4D + 5)y = 0$$

putting $m = D$

A.E $\Rightarrow m^2 - 4m + 5 = 0$

$\Rightarrow m^2$

Q.3. Solve;

$$y'' - 11y' + 30y = 0$$

$\Rightarrow$ Given;

$$y'' - 11y' + 30y = 0$$

$$\Rightarrow (D^2 - 11D + 30)y = 0$$

putting $m = D$

A.E $\Rightarrow m^2 - 11m + 30 = 0$

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$\Rightarrow m^2 - 6m - 5m + 30 = 0$

$\Rightarrow m(m-6) - 5(m-6) = 0$

$\Rightarrow (m-5)(m-6) = 0$

$\Rightarrow m = 5$ or $m = 6$

C.F $\Rightarrow y = C_1 e^{5x} + C_2 e^{6x}$.

Case-II :- If two roots are equal (i.e. $m_1 = m_2$),

C.F = only for two roots :

C.F $\Rightarrow y = C_1 x \cdot e^{mx} + C_2 e^{mx}$

- for nth root if two roots are equal :

C.F $\Rightarrow y = C_1 x \cdot e^{mx} + C_2 e^{mx} + \underbrace{C_3 e^{m_3 x} + \dots C_n e^{m_n x}}_{\text{rest are same as before}}$

- for nth root if three roots are equal :

C.F $\Rightarrow y = C_1 x^2 e^{mx} + C_2 x e^{mx} + C_3 e^{mx} + \underbrace{C_4 e^{m_4 x} + \dots C_n e^{m_n x}}_{\text{rest are same as before.}}$

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Q.4. Solve;

$$\frac{d^2x}{dt^2} + 6\frac{dx}{dt} + 9x = 0$$

⇒ Given;

$$\frac{d^2x}{dt^2} + 6\frac{dx}{dt} + 9x = 0$$

$$\Rightarrow (D^2 + 6D + 9)x = 0$$

putting $m = D$

$A.E \Rightarrow (m^2 + 6m + 9) = 0$

$\Rightarrow m^2 + 3m + 3m + 9 = 0$

$\Rightarrow m(m+3) + 3(m+3) = 0$

$\Rightarrow (m+3)(m+3) = 0$

$\Rightarrow m = -3, -3$

roots are same

$C.F \Rightarrow y = c_1 t e^{-3t} + c_2 e^{-3t}$

Q.5. Solve;

$$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 0$$

⇒ Given;

$$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 0$$

$$\Rightarrow (D^2 - 4D + 4)y = 0$$

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putting $m = D$

$$A.E \Rightarrow m^2 - 4m + 4 = 0$$
$$\Rightarrow m^2 - 2m - 2m + 4 = 0$$
$$\Rightarrow m(m-2) - 2(m-2) = 0$$
$$\Rightarrow (m-2)(m-2) = 0$$
$$\Rightarrow m = 2, 2$$

$$C.F \Rightarrow y = C_1 x e^{2x} + C_2 e^{2x}$$

Case - III :-

If one pair of roots be imaginary

i.e. $m_1 = \alpha + i\beta$
$m_2 = \alpha - i\beta$

$$C.F \Rightarrow y = e^{\alpha x}[C_1 \cos\beta x + C_2 \sin\beta x] + \underbrace{C_3 e^{m_3 x} + \dots C_n e^{m_n x}}_{\text{same as before}}$$

$\downarrow$

No use only proof.

$$y = C_1 e^{(\alpha + i\beta)x} + C_2 e^{(\alpha - i\beta)x}$$
$$= C_1 e^{\alpha x} \times e^{i\beta x} + C_2 e^{\alpha x} \times e^{-i\beta x}$$
$$= e^{\alpha x}[C_1 e^{i\beta x} + C_2 e^{-i\beta x}]$$
$$[e^{ni\theta} = \cos n\theta + i \sin n\theta]$$
$$= e^{\alpha x}[C_1(\cos\beta x + i \sin\beta x) + C_2(\cos\beta x - \sin\beta x)]$$
$$= e^{\alpha x}[\cos\beta x(C_1 + C_2) + i \sin\beta x(C_1 - C_2)]$$
$$= e^{\alpha x}[C_1 \cos\beta + C_2 \sin\beta x], \quad C_1 = C_1 + C_2, \quad C_2 = i(C_1 - C_2),$$

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Q.6. Solve;

$$(D^3+D^2+4D+4)\,y=0$$

→ Given;

$$(D^3+D^2+4D+4)\,y=0$$

putting $m=D$

A.E ⇒ $m^3+m^2+4m+4=0$

$\Rightarrow m^2[m+1]+4[m+1]=0$

$\Rightarrow [m+1][m^2+4]=0$

$\Rightarrow m=-1$ or $m=2i$ or $m=-2i$

C.F ⇒ $y=C_1e^{-x}+C_2e^{2ix}+C_3e^{-2ix}$

$$y=C_1e^{-x}+C_2\cos 2x+C_3\sin 2x$$

Q.7. Solve; $y''-2y'+5y=0$

⇒ Given;

$$y''-2y'+5y=0$$

$$\Rightarrow (D^2-2D+5)\,y=0$$

putting $m=D$

A.E ⇒ $m^2-2m+5=0$

$$\Rightarrow m=\frac{2\pm\sqrt{4-20}}{2}$$

$$=\frac{2\pm\sqrt{-16}}{2}=\frac{2\pm 4i}{2}=1\pm 2i$$

$\Rightarrow m=1+2i,\ 1-2i$

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$$C.F \Rightarrow y = C_1 e^{(1+2i)x} + C_2 e^{(1-2i)x}$$

$$= C_1(e^x \cdot e^{2ix}) + C_2(e^x \cdot e^{-2ix})$$

$$= e^x \left[C_1 e^{2ix} + C_2 e^{-2ix} \right]$$

$$y = e^x \left[C_1 \cos 2x + C_2 \sin 2x \right]$$

Q.8. Solve;

$$\frac{d^2x}{dt^2} - 4\frac{dx}{dt} + 13x = 0$$

$\Rightarrow$ Given;

$$\frac{d^2x}{dt^2} - 4\frac{dx}{dt} + 13x = 0$$

$$\Rightarrow (D^2 - 4D + 13)x = 0$$

putting $m = D$

$$A.E \Rightarrow m^2 - 4m + 13 = 0$$

$$\Rightarrow m = \frac{4 \pm \sqrt{16-52}}{2}$$

$$= \frac{4 \pm \sqrt{-36}}{2}$$

$$= \frac{4 \pm 6i}{2} = 2 \pm 3i$$

$$\Rightarrow m = 2+3i,\ 2-3i$$

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$$\cancel{C.F \Rightarrow c_1 e^{(2+3i)x} + c_2 e^{(2-3i)x}}$$

$$\cancel{y = c_1(e^{2x} \cdot e^{3ix}) + c_2(e^{2x} \cdot e^{-3ix})}$$

$$\cancel{y =}$$

$$C.F \Rightarrow x = c_1 e^{(2+3i)t} + c_2 e^{(2-3i)t}$$

$$= c_1(e^{2t} \cdot e^{3it}) + c_2(e^{2t} \cdot e^{-3it})$$

$$\cancel{= e^{2t}(c_1 + c_2)}$$

$$= e^{2t}[c_1 e^{3it} + c_2 e^{-3it}]$$

$$x = e^{2t}[c_1 \cos 3t + c_2 \sin 3t]$$

Q.9. $(D^3 - 3D^2 + D - 3)y = 0$

$\Rightarrow$ Given;

$$(D^3 - 3D^2 + D - 3)y = 0$$

putting $m = D$

$$A.E \Rightarrow (m^3 - 3m^2 + m - 3) = 0$$

$$\Rightarrow m^2[m-3] + 1[m-3] = 0$$

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$\Rightarrow [m-3][m^2+1] = 0$

$\Rightarrow m = 3, m = i, m = -i$

C.F $\Rightarrow$ $y = C_1 e^{+3x} + C_2 e^{ix} + C_3 e^{-ix}$

$y = C_1 e^{3x} + C_2 \cos x + C_3 \sin x$

Case-IV :- If two points of imaginary roots be equal i.e.

$m_1 = m_2 = \alpha + i\beta$
$m_3 = m_4 = \alpha - i\beta$

C.F $\Rightarrow$

$y = e^{\alpha x}[(C_1 x + C_2)\cos\beta x + (C_3 x + C_4)\sin\beta x] + \cdots$

- If three point of imaginary roots are equal i.e,

$m_1 = m_2 = m_3 = \alpha + i\beta$
$m_4 = m_5 = m_6 = \alpha - i\beta$

C.F $\Rightarrow$

$y = e^{\alpha x}[(C_1 x^2 + C_2 x + C_3)\cos\beta x + (C_4 x^2 + C_5 x + C_6 x)\sin\beta x]$
$+ \cdots$

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Q.10 Solve;

$$\frac{d^2y}{dx^2} + y = 0$$

=> Given;

$$\frac{d^2y}{dx^2} + y = 0$$

$$\Rightarrow (D^2+1)y = 0$$

putting $m = D$

A.E => $m^2+1 = 0$

$m = i, -i$

Same imaginary root

C.F => ~~$y = C_1 e^{ix} + C_2 e^{-ix}$~~

$$y = e^{\alpha x}[(C_1x + C_2)\cos\beta x + (C_3x + C_4)\sin\beta x]$$

$$y = e^{0}[(C_1x + C_2)\cos x + (C_3x + C_4)\sin x]$$

$$y = C_1x\cos x + C_3x\sin x + C_2\cos x + C_4\sin x$$

Q.11 Solve;

$$(D^2+1)^3y = 0.$$

=> Given;

$$(D^2+1)^3y = 0$$

putting $m = D$

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